



## Pre-De-Icer – Ingredient Cost Analysis

### Production Cost Estimate (1 Gallon Batch)

Target Batch Volume: 1 U.S. Gallon (3,785 mL / 128 fl oz)

### Ingredient Breakdown – Scaled from Base Formula

- **Methylcellulose (food-grade, finely powdered)**
  - 103 teaspoons (approx. 1.2 ounces)
  - Cost per ounce: \$1.25
  - Total: \$1.50
- **Isopropyl Alcohol (70–91%) 91% alcohol is higher cost than 70%**
  - 2,065 mL (approx. 69.7 fl oz)
  - Cost per gallon: \$17.50
  - Cost per ounce: \$0.137
  - Total: \$9.55
- **Water (distilled or tap, as needed) Distilled would cost to make or purchase~cheaper to make own distilled in the long run**
  - 1,376 mL (approx. 46.6 fl oz)
  - Cost: \$0.00
- **Propylene Glycol (corn based)**
  - 344 mL (approx. 11.6 fl oz)
  - Cost per gallon: \$20.00
  - Cost per ounce: \$0.156

– Total: \$1.81

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**Total Cost per Gallon: \$12.86**

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### Cost per 16 oz Retail Bottle

- $128 \text{ oz} \div 16 \text{ oz} = 8 \text{ bottles per gallon}$
  - $\$12.86 \div 8 = \$1.61 \text{ per bottle (ingredients only)}$
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### Key Notes

- Ingredient prices based on small-batch supply costs. Bulk purchasing could reduce total cost by 20–40%.
- Water assumed as no-cost input (filtered or tap).
- Does not include bottling, packaging, labeling, shipping, labor, or overhead.
- Methylcellulose density approx. 85 teaspoons per ounce, based on fine powder granule size.

